Lecture - 4

Chapter 2

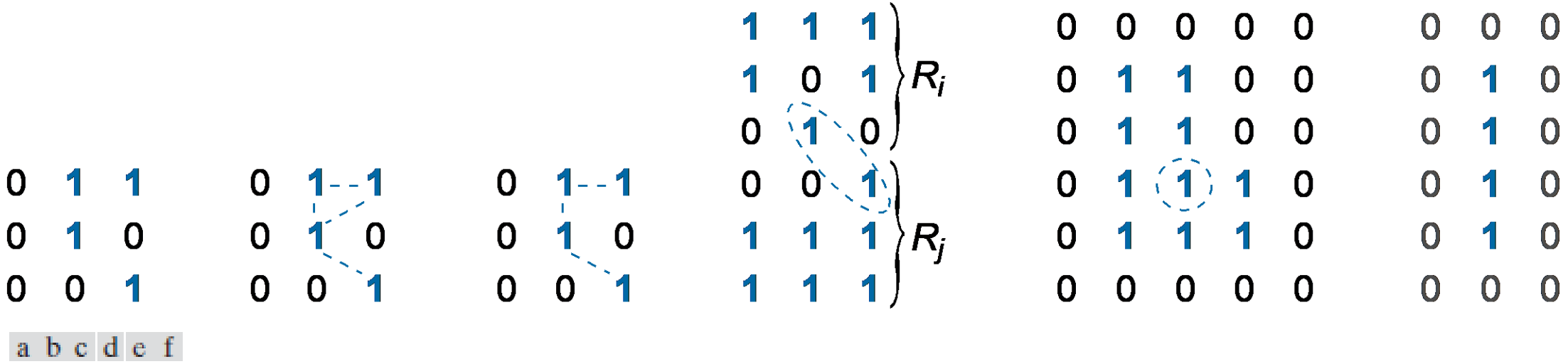
DIGITAL IMAGE FUNDAMENTALS

Pixel Adjacency

$$N_4 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad N_D = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad N_8 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

- Adjacency:
  - p and q are 4-adjacent: $q \in N_4(p)$
  - p and q are 8-adjacent: $q \in N_8(p)$
  - p and q are m-adjacent: $\{q \in N_4(p)\} or \{\{q \in N_D(p)\} and \{N_4(p) \cap N_4(q)\} = \varnothing\}$

Pixel Adjacency



a b c d e f

FIGURE 2.28 (a) An arrangement of pixels. (b) Pixels that are 8-adjacent (adjacency is shown by dashed lines). (c) *m*-adjacency. (d) Two regions (of 1's) that are 8-adjacent. (e) The circled point is on the boundary of the 1-valued pixels only if 8-adjacency between the region and background is used. (f) The inner boundary of the 1-valued region does not form a closed path, but its outer boundary does.

Distance between Pixels

- Euclidean: $D_e(p,q) = \sqrt{(x-s)^2 + (y-t)^2}$
- D_4 (City Block or Manhattan): $D_4(p,q) = |x-s| + |y-t|$
- D_8 (Chessboard): $D_8(p,q) = Max\{|x-s|, |y-t|\}$

$$\begin{array}{ccccc} & & 2 & & \\ & 2 & 1 & 2 & \\ 2 & 1 & 0 & 1 & 2 \\ & 2 & 1 & 2 & \\ & & 2 & & \end{array}$$

- Constant Distance Contour:
  - D4 (Left)
  - D8 (Right)

$$\begin{array}{ccccc} 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 1 & 1 & 2 \\ 2 & 1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 1 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{array}$$

Pixel Connectivity

Connectivity forms the basis for establishing the boundaries of an objects and also components of regions in an image. **Two pixels are connected if** :

1. if the pixels are adjacent (e.g., they are 4-neighbours)
2. if their gray levels satisfy a specified criterion of similarity. For example, in a binary image with values 0 and 1, two-pixels may be 4-neighbors, but they are said to be connected only if they have the same value (Pixels in grayscale image will be connected if they belongs to a set V defining grayscale range e.g., V={50-100})

Connected Set - S

Two pixels are said to be connected in S (subset of pixels in an image) if there exists a path between them consisting entirely of pixels in S

For any pixel p in S, the set of pixels that are connected to it in S is called a connected component of S.

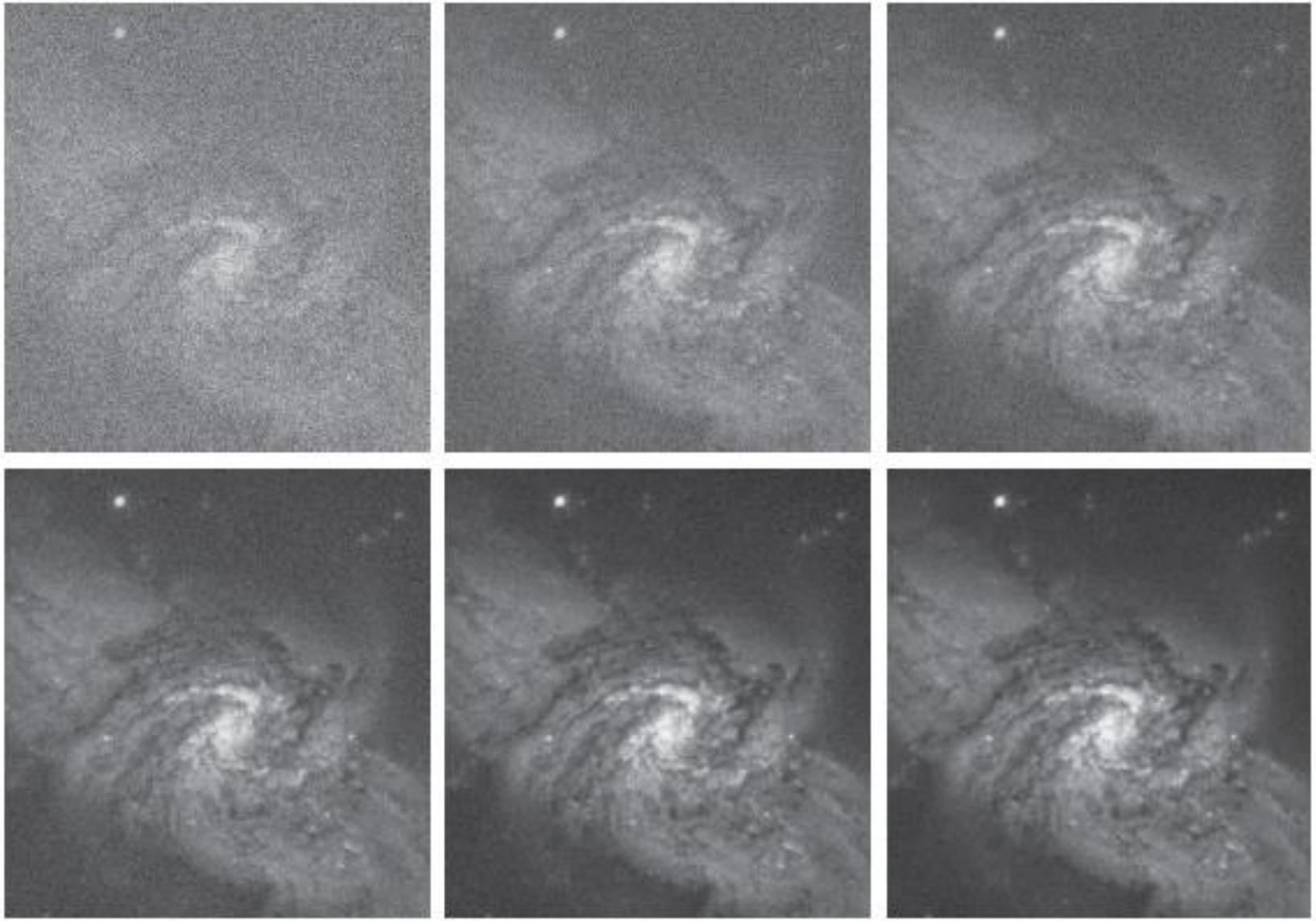
If it only has one connected component, then set S is called a connected set.

Region - R

R is a region of the image if R is a connected set.

The boundary (border or contour) of a region R is the set of pixels in the region that have one or more neighbors that are not in R.

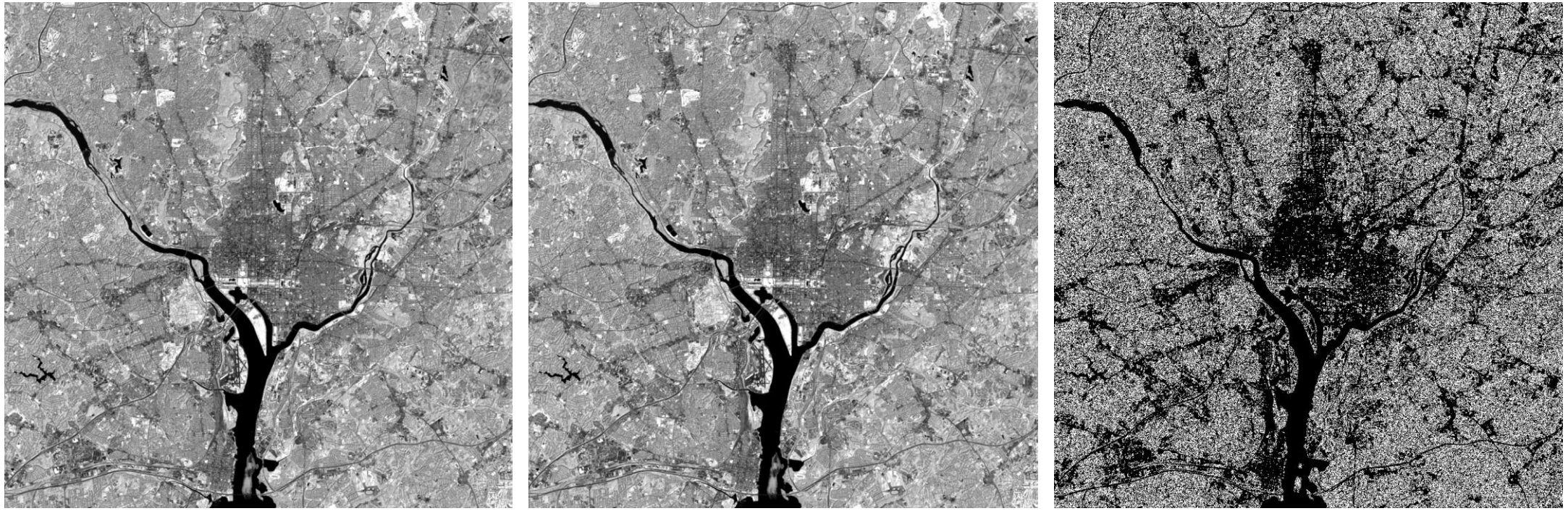
Image Averaging



a b c
d e f

FIGURE 2.29 (a) Image of Galaxy Pair NGC 3314 corrupted by additive Gaussian noise. (b)-(f) Result of averaging 5, 10, 20, 50, and 1,00 noisy images, respectively. All images are of size 566×598 pixels, and all were scaled so that their intensities would span the full [0, 255] intensity scale. (Original image courtesy of NASA.)

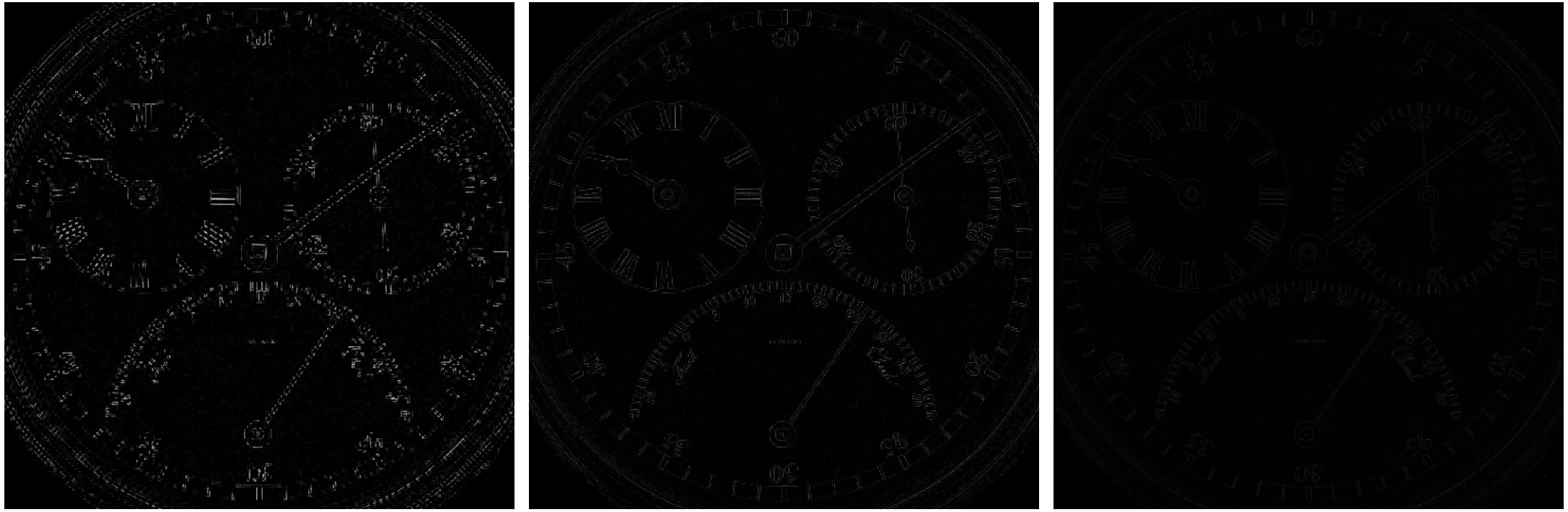
Image Subtraction



a b c

FIGURE 2.30 (a) Infrared image of the Washington, D.C. area. (b) Image resulting from setting to zero the least significant bit of every pixel in (a). (c) Difference of the two images, scaled to the range [0, 255] for clarity. (Original image courtesy of NASA.)

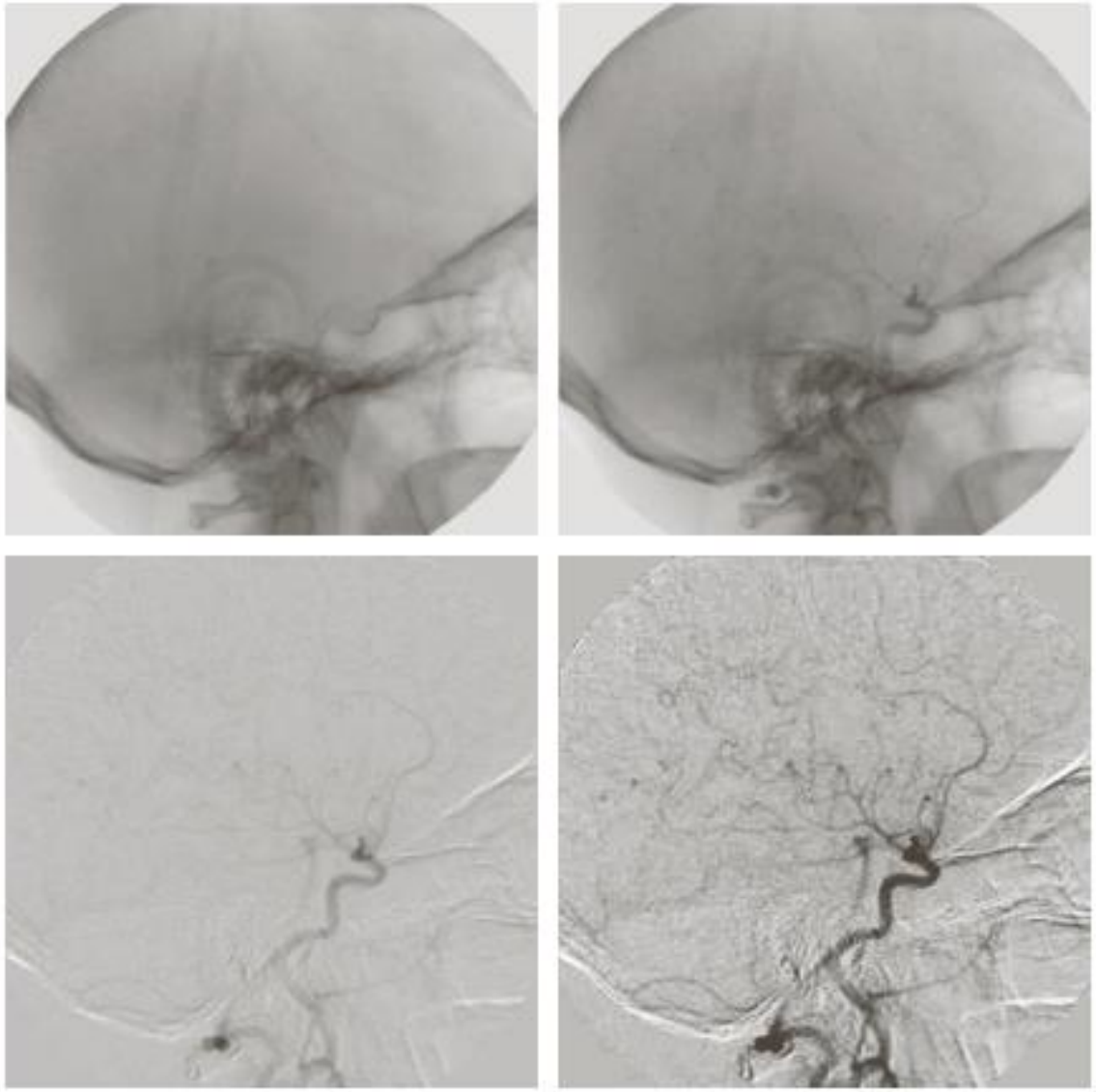
Image Subtraction



a b c

FIGURE 2.31 (a) Difference between the 930 dpi and 72 dpi images in Fig. 2.23. (b) Difference between the 930 dpi and 150 dpi images. (c) Difference between the 930 dpi and 300 dpi images.

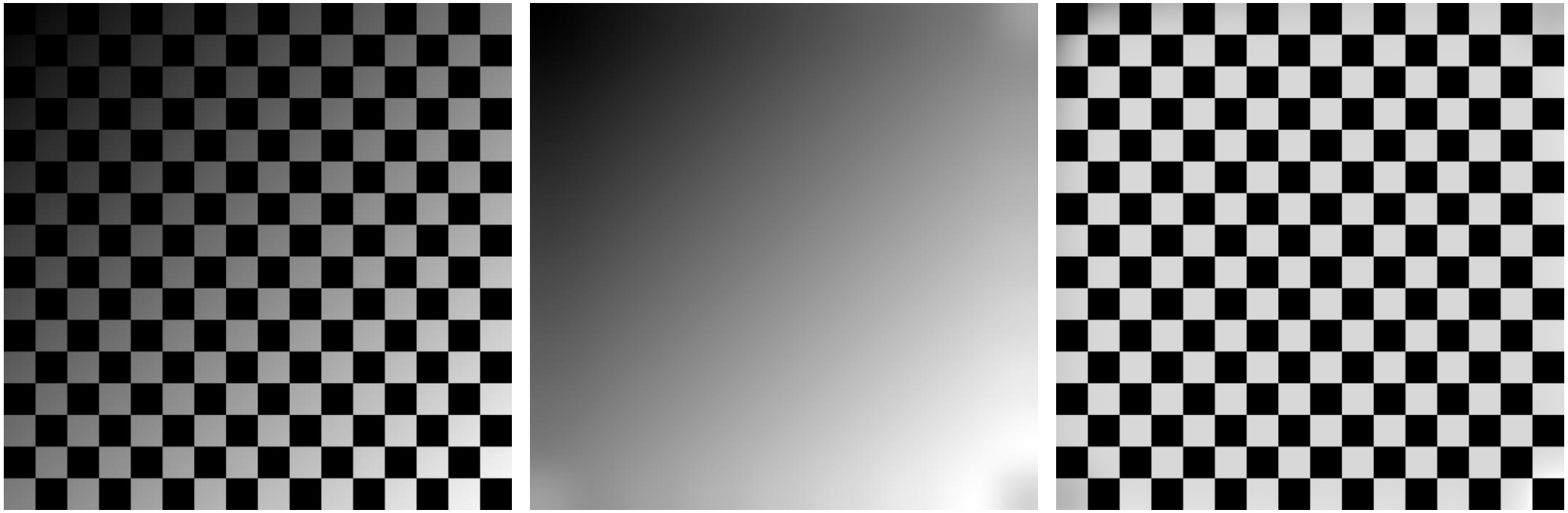
Image Subtraction



a b
c d

FIGURE 2.32 Digital subtraction angiography. (a) Mask image. (b) A live image. (c) Difference between (a) and (b). (d) Enhanced difference image. (Figures (a) and (b) courtesy of the Image Sciences Institute, University Medical Center, Utrecht, The Netherlands.)

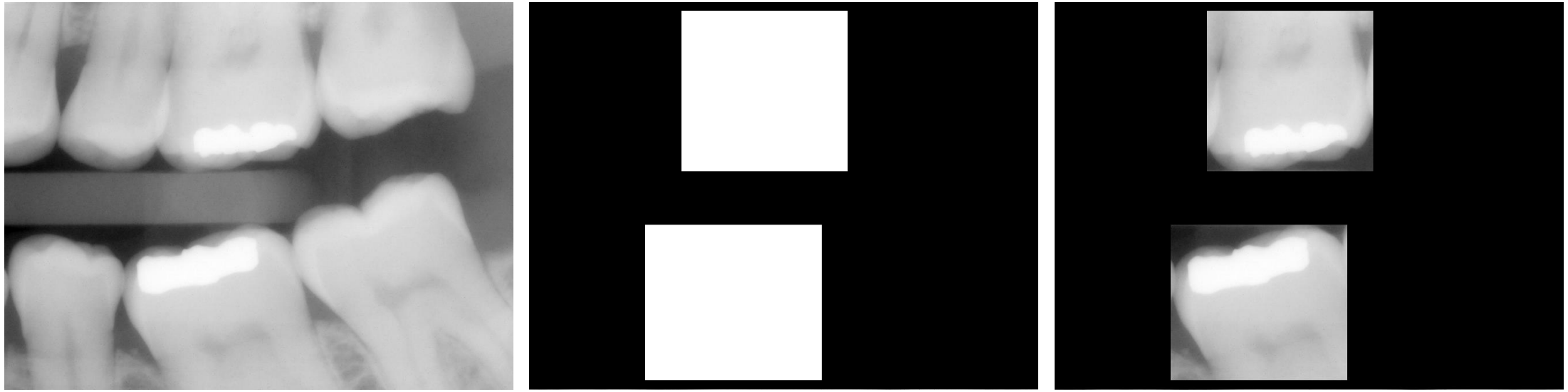
Shading Correction



a b c

FIGURE 2.33 Shading correction. (a) Shaded test pattern. (b) Estimated shading pattern. (c) Product of (a) by the reciprocal of (b). (See Section 3.5 for a discussion of how (b) was estimated.)

Region of Interest (RoI) Masking



a b c

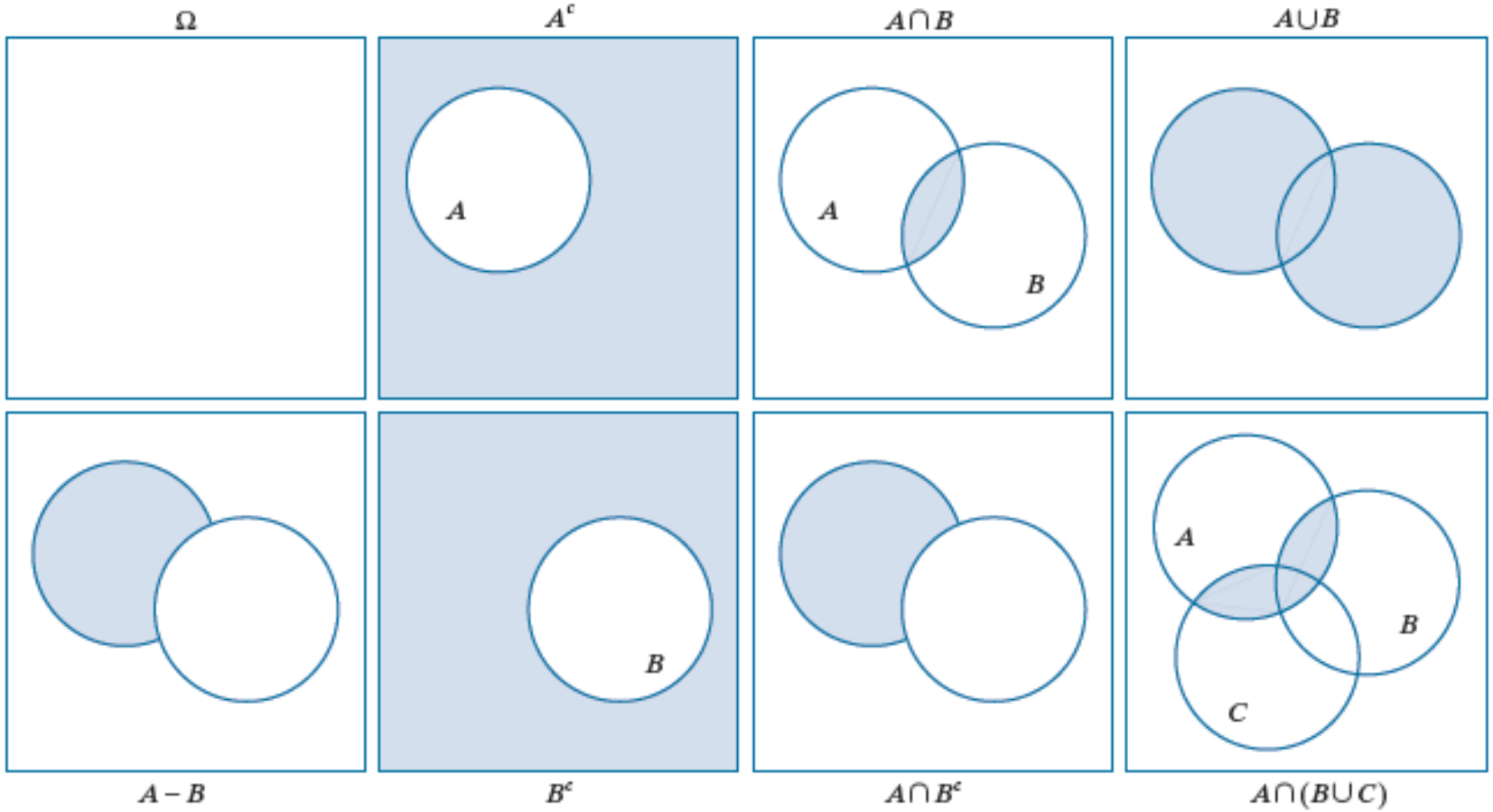
FIGURE 2.34 (a) Digital dental X-ray image. (b) ROI mask for isolating teeth with fillings (white corresponds to 1 and black corresponds to 0). (c) Product of (a) and (b).

SET Operations

TABLE 2.1
Some important set operations and relationships.

Description	Expressions
Operations between the sample space and null sets	$\Omega^c = \varnothing;\ \varnothing^c = \Omega;\ \Omega \cup \varnothing = \Omega;\ \Omega \cap \varnothing = \varnothing$
Union and intersection with the null and sample space sets	$A \cup \varnothing = A;\ A \cap \varnothing = \varnothing;\ A \cup \Omega = \Omega;\ A \cap \Omega = A$
Union and intersection of a set with itself	$A \cup A = A;\ A \cap A = A$
Union and intersection of a set with its complement	$A \cup A^c = \Omega;\ A \cap A^c = \varnothing$
Commutative laws	$A \cup B = B \cup A$ $A \cap B = B \cap A$
Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distributive laws	$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$
DeMorgan's laws	$(A \cup B)^c = A^c \cap B^c$ $(A \cap B)^c = A^c \cup B^c$

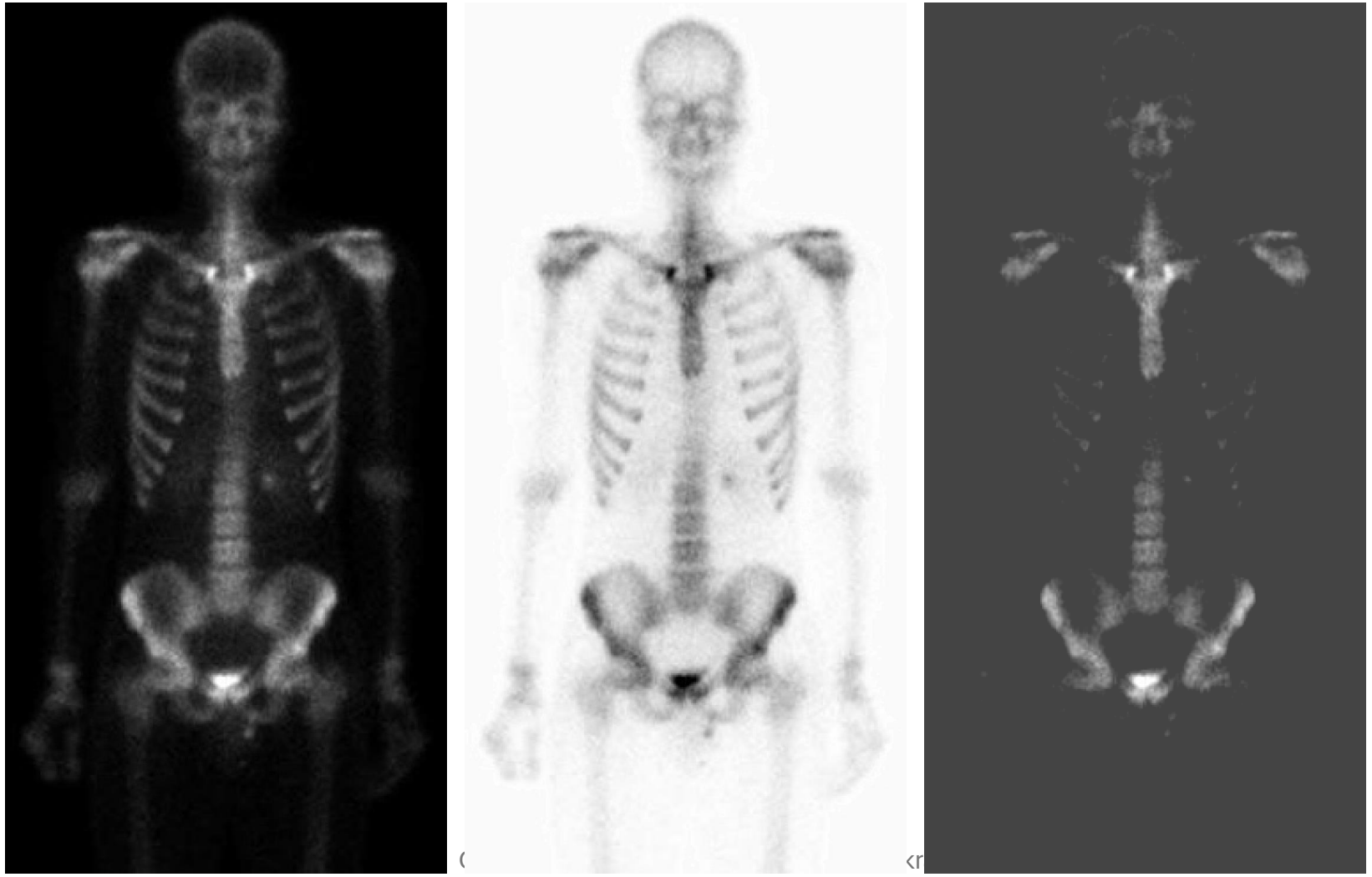
SET Operations



a b c d
e f g h

FIGURE 2.35 Venn diagrams corresponding to some of the set operations in Table 2.1. The results of the operations, such as A^c, are shown shaded. Figures (e) and (g) are the same, proving via Venn diagrams that $A - B = A \cap B^c$ [see Eq. (2-40)].

SET Operations on Grayscale Images



a b c

FIGURE 2.36 Set operations involving grayscale images. (a) Original image. (b) Image negative obtained using grayscale set complementation. (c) The union of image (a) and a constant image. (Original image courtesy of G.E. Medical Systems.)

Logical Operations

TABLE 2.2 Truth table defining the logical operators AND(∧), OR(∨), and NOT(~).

a	b	a AND b	a OR b	NOT(a)
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

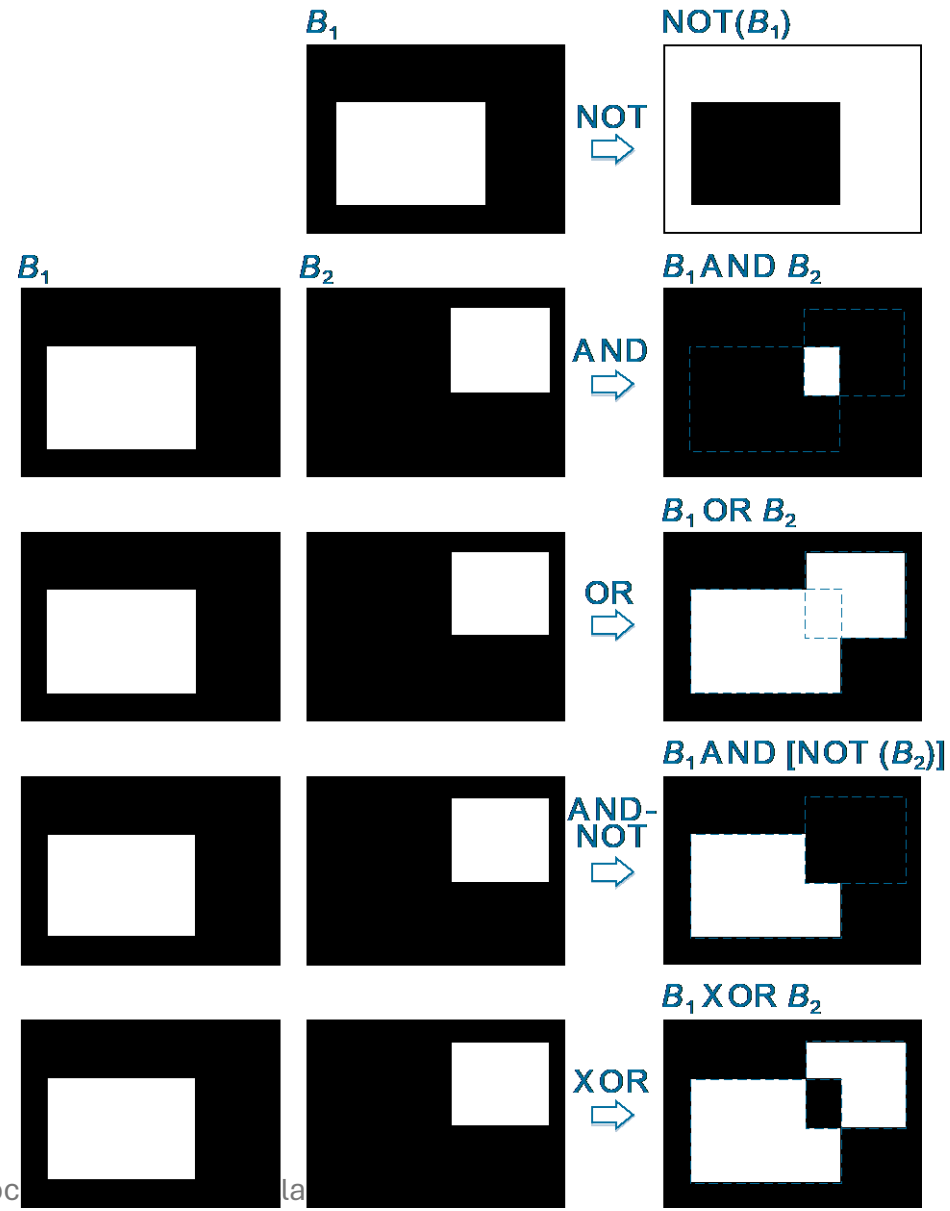
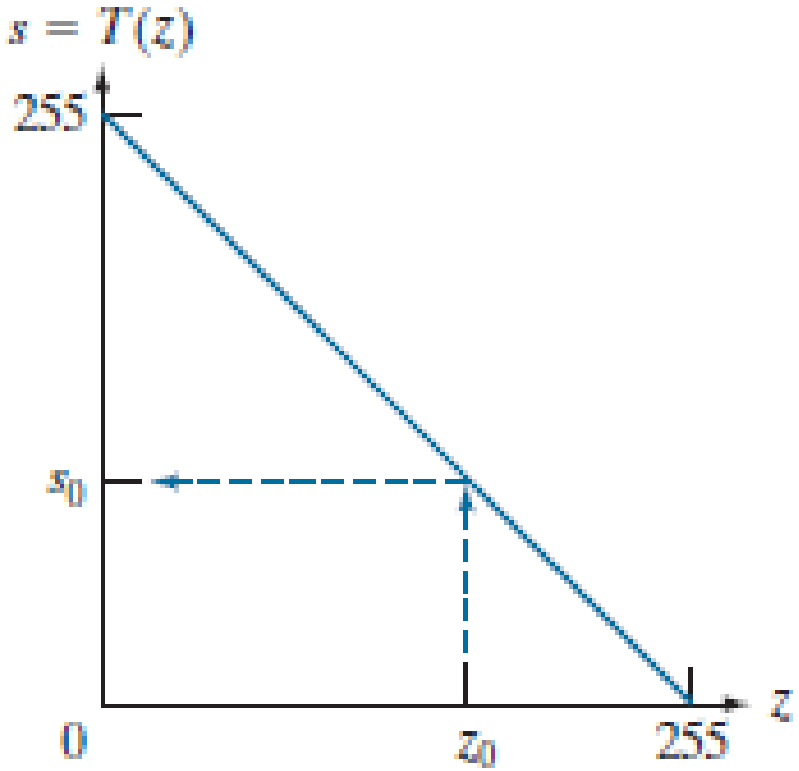


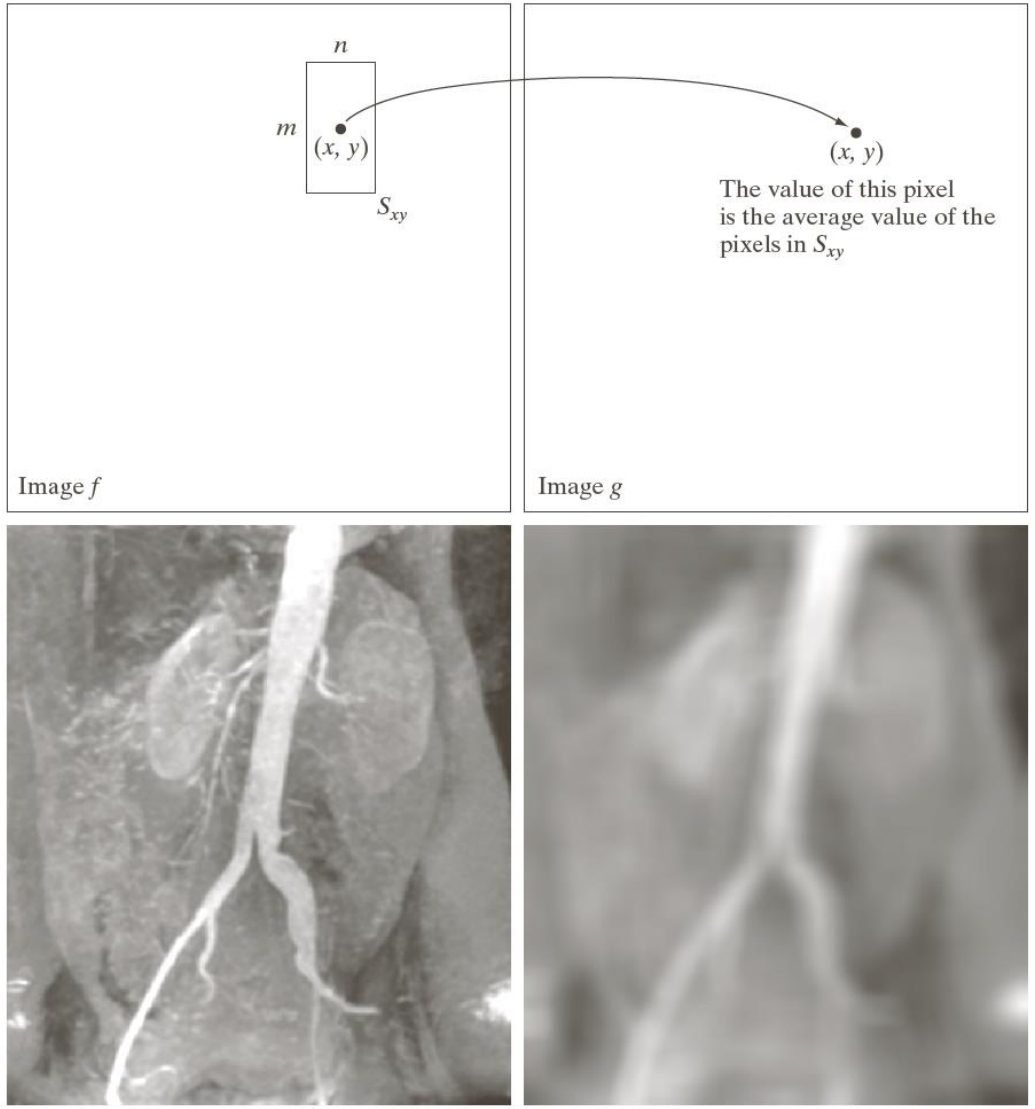
FIGURE 2.37 Illustration of logical operations involving foreground (white) pixels. Black represents binary 0's and white binary 1's. The dashed lines are shown for reference only. They are not part of the result.

Pixel Processing (Intensity Transformation)

FIGURE 2.38 Intensity transformation function used to obtain the digital equivalent of photographic negative of an 8-bit image..



Neighborhood Processing (Local Averaging)



a b
c d

FIGURE 2.39 Local averaging using neighborhood processing. The procedure is illustrated in (a) and (b) for a rectangular neighborhood. (c) An aortic angiogram (see Section 1.3). (d) The result of using Eq. (2-43) with $m = n = 41$. The images are of size 790×686 pixels. (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

Image Geometric Transformation (Affine)

- Geometric Spatial Transform
  - General Formulation:

$$(x, y) = T\{(u, v)\}$$

  - Example (Affine Transform)

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} u \\ v \\ 1 \end{pmatrix}$$

TABLE 2.3 Affine transformations based on Eq. (2-45).

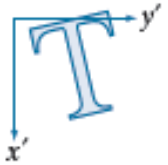
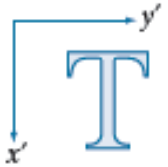
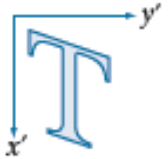
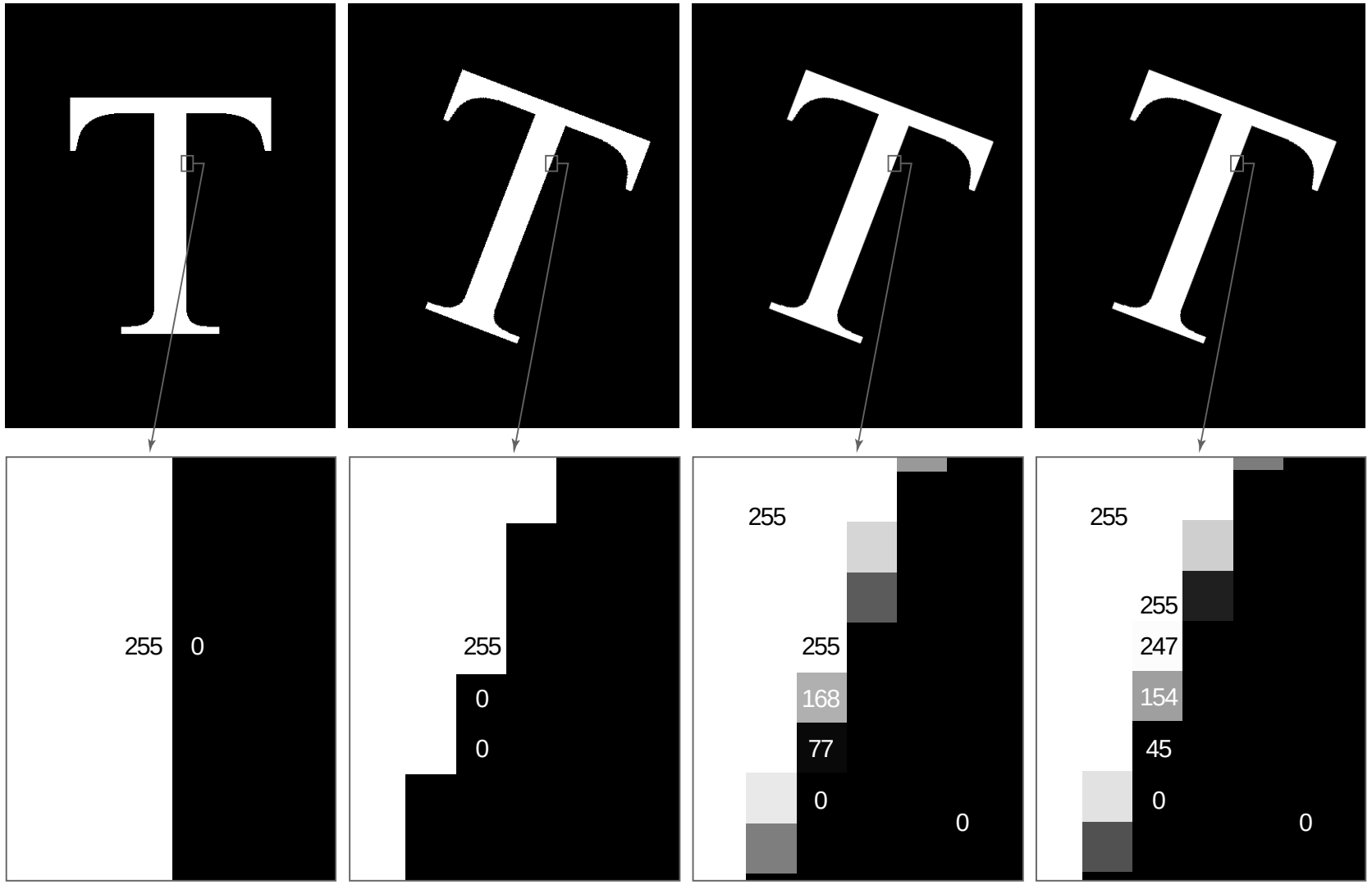
Transformation Name	Affine Matrix, A	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = y$	
Scaling/Reflection (For reflection, set one scaling factor to −1 and the other to 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = c_x x$ $y' = c_y y$	
Rotation (about the origin)	$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x\cos\theta - y\sin\theta$ $y' = x\sin\theta + y\cos\theta$	
Translation	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + t_x$ $y' = y + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + s_v y$ $y' = y$	
Shear (horizontal)	$\begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = s_h x + y$	

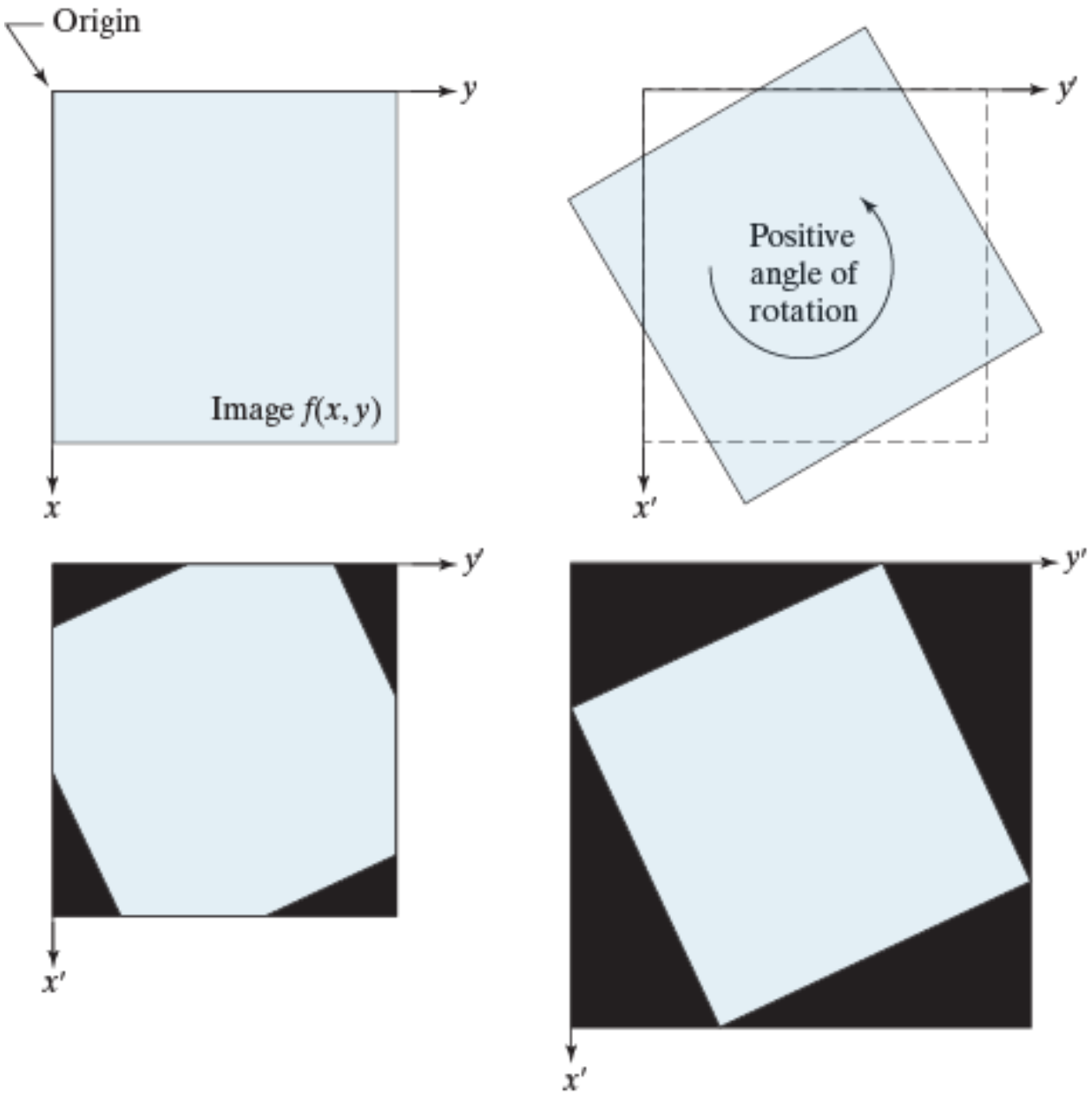
Image Rotation



a b c d
e f g h

FIGURE 2.40 (a) A 541 × 421 image of the letter T. (b) Image rotated –21° using nearest-neighbor interpolation for intensity assignments. (c) Image rotated –21° using bilinear interpolation. (d) Image rotated –21° using bicubic interpolation. (e)-(h) Zoomed sections (each square is one pixel, and the numbers shown are intensity values).

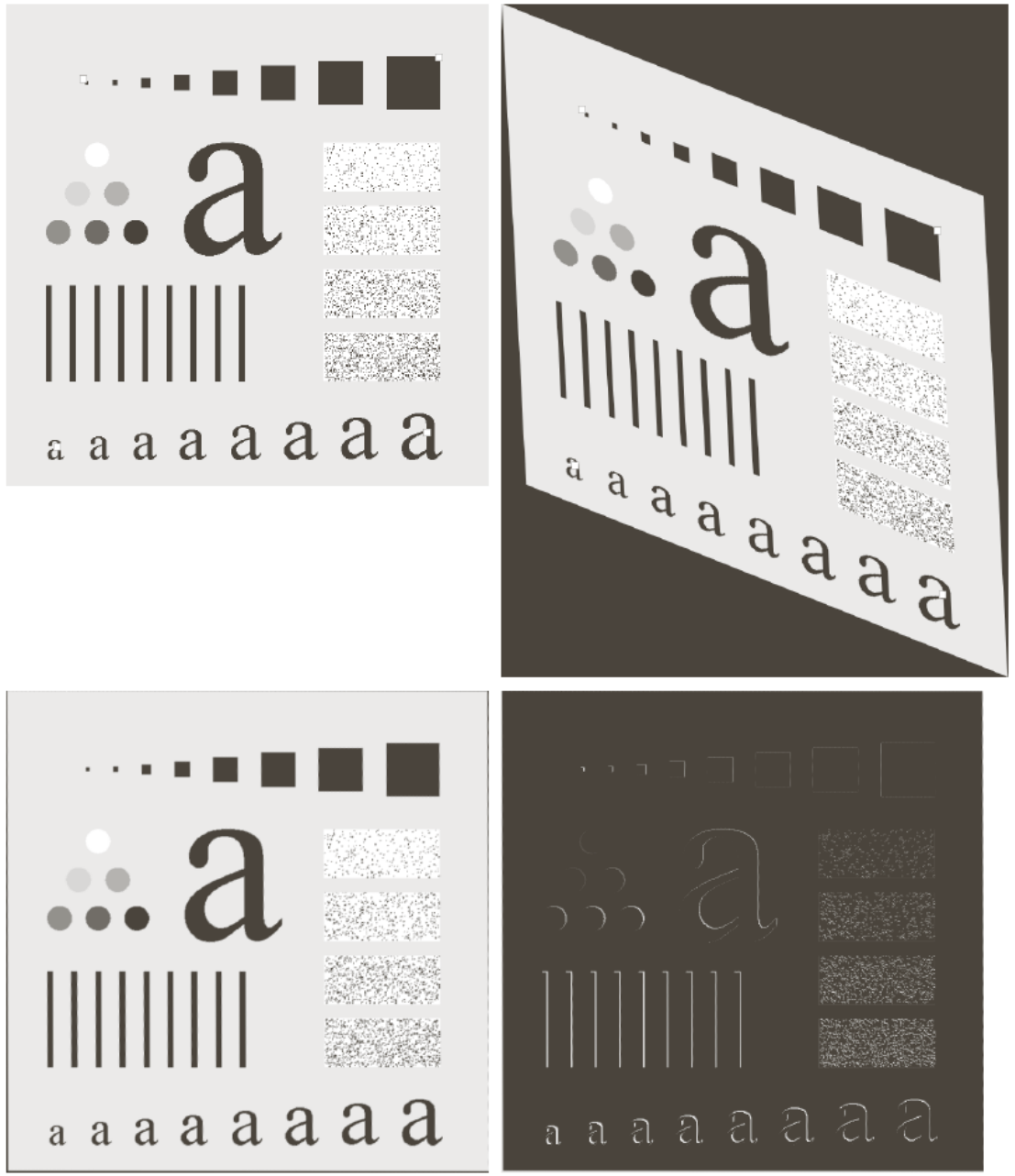
Image Rotation



a b
c d

FIGURE 2.41 (a) A digital image. (b) Rotated image (note the counterclockwise direction for a positive angle of rotation). (c) Rotated image cropped to fit the same area as the original image. (d) Image enlarged to accommodate the entire rotated image.

Image Registration



a b
c d

FIGURE 2.42 Image registration. (a) Reference image. (b) Input (geometrically distorted image). Corresponding tie points are shown as small white squares near the corners. (c) Registered (output) image (note the errors in the border). (d) Difference between (a) and (c), showing more registration errors.

Pixel Intensity Vector of Color Image

FIGURE 2.43
Forming a vector from corresponding pixel values in three RGB component images.

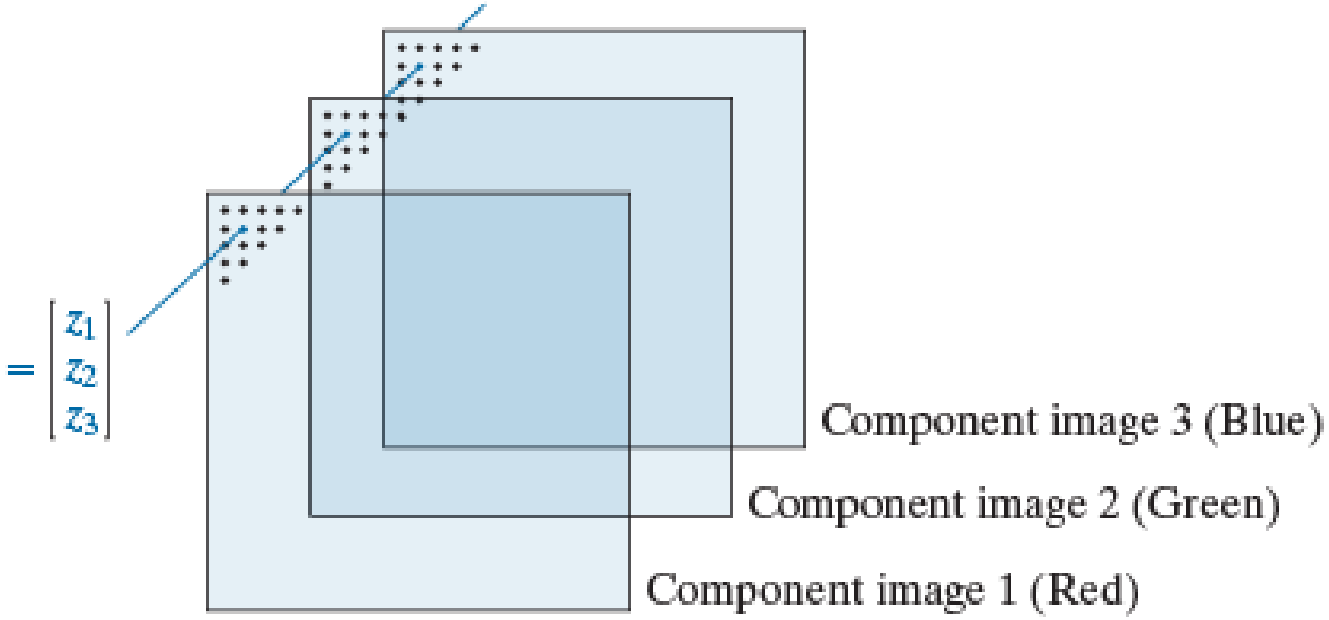
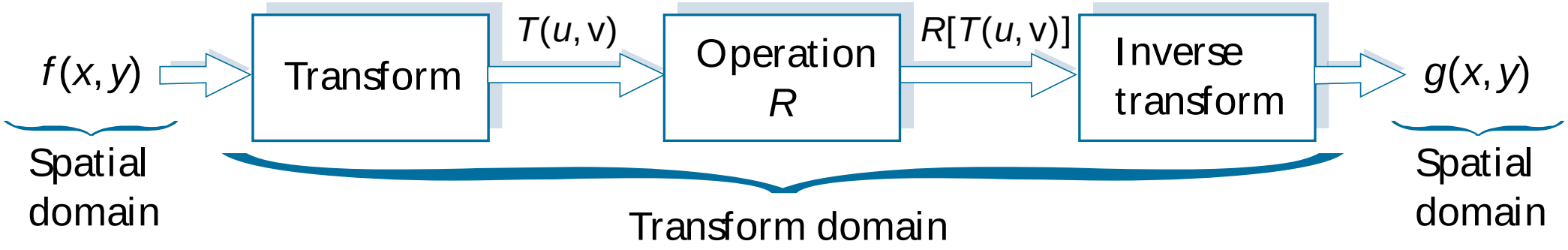
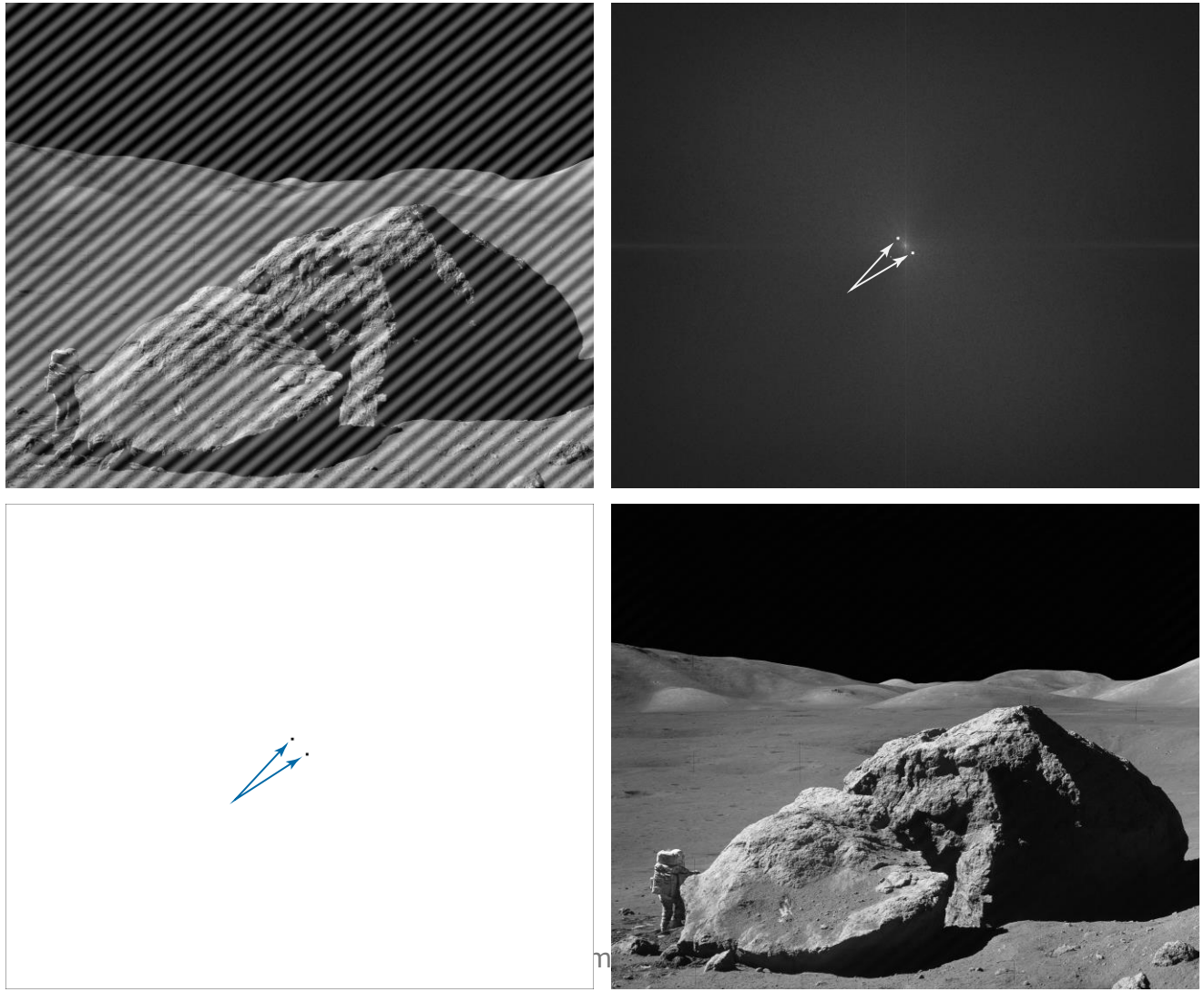


Image Transformation

FIGURE 2.44 General approach for working in the linear transform domain.



Sinusoidal Noise Removal (frequency domain)



a b
c d

FIGURE 2.45 (a) Image corrupted by sinusoidal interference. (b) Magnitude of the Fourier transform showing the bursts of energy caused by the interference (the bursts were enlarged for display purposes). (c) Mask used to eliminate the energy bursts. (d) Result of computing the inverse of the modified Fourier transform. (Original image courtesy of NASA.)